Modelling of Significant Wave Height Time Series Prediction Using CatBoost With A Case Study in Pacitan, Indonesia

1st Osa Nastiyar Maulani
School of Computing
Telkom University
Bandung, Indonesia
osanastiyar@student.telkomuniversity.ac.id

2nd Didit Adytia
School of Computing
Telkom University
Bandung, Indonesia
aditya@telkomuniversity.ac.id

3rd Hendri Bhirowo Dwi Hananto
General Division
PLN Research Institute
Jakarta, Indonesia
hendri.bhirowo@pln.co.id

4th Indra.A.Aditya
General Divison
PLN Research Institute
Jakarta, Indonesia
indra.aditya@pln.co.id

2025 International Conference on Advancement in Data Science, E-learning and Information System (ICADEIS) | 979-8-3315-1332-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICADEIS65852.2025.10933175

***Abstract*— Forecasting wave height is essential for managing coastal activities, as it mitigates risks and losses in maritime transportation and informs coastal infrastructure design. Numerical wave modeling is frequently employed in forecasting wave heights. However, this methodology requires significant computational time and high grid resolution to obtain more accurate forecasts. Meanwhile, statistical approaches employing trend analysis of historical data for wave height prediction exhibit limitations in addressing extreme scenarios and characterizing intricate nonlinear interactions. This work proposes using a machine learning model called CatBoost to predict wave height time series. We choose high-resolution wave data from nested wave simulations utilizing the SWAN model at Pacitan Beach, East Java, Indonesia. We forecasted for intervals of 1, 5, 7, and 14 days, utilizing diverse durations of wave data. Furthermore, the Catboost model's efficacy is evaluated compared to other boosting models, including Adaboost and XGboost. We conducted hyperparameter optimization with GridSearch to get optimal performance with Catboost. The results demonstrated that the CatBoost model attained the maximum accuracy in all forecasting situations, with an RMSE of 0.0121 and an R2 value of 0.9986 over a 14-day prediction interval.**

Keywords*— *AdaBoost, CatBoost, Hyperparameter Tuning, Wave Height Forecasting, XGBoost

I. INTRODUCTION

Ocean waves are an important natural phenomenon that affects offshore and coastal operations. Wave height analysis is essential to ensure navigation safety, optimise marine operations, and reduce the hazards of natural disasters. Precise wave height forecasts can increase production, save fuel and prevent dangerous scenarios. Individuals in marine and coastal cities can benefit from systems capable of forecasting wave patterns around the clock. However, ocean waves' complex and erratic nature makes it challenging to calculate wave height [1].

Several methods exist for precisely forecasting wave heights. Nonetheless, several issues remain unresolved. Numerical approaches cannot deliver intricate wave representations and need substantial processing resources [2]. Conversely, statistical methodologies employing trend analysis of historical data have been utilised for projecting wave heights. However, these statistical approaches are limited to handling complex scenarios and cannot describe complicated nonlinear interactions [3]. Therefore, machine learning becomes a very effective way to numerically and statistically improve wave forecasting models. This approach was chosen because it can optimise large datasets for accurate predictions. Its effectiveness has been proven and is now used in modern forecasting methods [4]. For example, Anggraeni et al. [2] Conducted research comparing the AdaBoost and XGBoost methodologies for forecasting wave heights across periods of seven, 14, 30, 45, and 60 days in the Pangandaran area of Indonesia. The research employed five years of training data with four, five, and six months of experimental data. Their investigation demonstrated that XGBoost yielded superior results compared to AdaBoost; XGBoost achieved a predictive evaluation metric RMSE of 0.093 and a correlation coefficient (CC) of 0.989, whereas AdaBoost attained an RMSE of 0.110 and a CC of 0.989. XGBoost and AdaBoost, while widely utilised algorithms possess inherent restrictions. XGBoost is prone to overfitting with noisy data, but AdaBoost is suboptimal for time series data characteristics [5], [6]. Elsworth et al. [7] Also Executed a study with the LSTM model. The LSTM model attains an overall average score of 88.39 across the all-time series, with an average calculation duration of 1293 seconds per time series. The significant calculation time, although precise, is a substantial limitation due to the complexity of optimizing the LSTM model parameters.

This research uses a machine learning called the CatBoost model to forecast wave data and seeks to improve resolution across networks worldwide. CatBoost is intended to provide superior performance and fast training duration to handle massive data sets while maintaining efficiency [8]. Additionally, the model employs strategies to reduce the risk of overfitting, thereby increasing its capacity to generalise previously unseen data [9]. This study uses CatBoost to handle wave height prediction in a region affected by significant ocean waves. In this paper, we examine the coastal area of Pacitan Beach in East Java, Indonesia, which is subject to considerable wave and surge action from the Indian Ocean. We performed high-resolution nested wave simulations utilising the SWAN model to get precise wave data for the Pacitan coastal region[1], [10], [11]. We assessed the efficacy of the CatBoost model in predicting wave heights 1, 5, 7, and 14 days in advance across various situations with differing training data durations over periods of 1, 2, and 8 years. The CatBoost model was optimised for hyperparameters using the GridSearch approach. Furthermore, its efficacy was compared to other prevalent machine learning models for time series forecasting, such as XGBoost and AdaBoost. Root Mean Square Error (RMSE) and R-squared (R^2) measures evaluate the model's efficacy.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.

This paper's structure is as follows: Section II discusses the literature review on wave height forecasting, CatBoost, XGBoost, and AdaBoost. Section III describes the methodology used. Section IV analyses the forecasting outcomes obtained from XGBoost and AdaBoost. Section V presents the conclusion of this study.

II. Literature Review

A. Wave Height Forecasting

Wave height forecasting estimates the amplitude of ocean waves, utilizing historical data, meteorological patterns, and environmental factors. Numerical approaches exhibit constraints in resolution and computing demands [2]. Statistical methods are less efficacious in addressing extreme values and nonlinear interactions [3]. Machine learning has lately demonstrated significant potential in enhancing predicting accuracy by managing large and intricate data sets. It has proven useful and utilized for many forecasting challenges [4]. In 2020, Fan et al. employed SWAN-LSTM for precise wave forecasting at one and six-hour intervals. [12]. Adytia et al. utilized GRNN in the same year, achieving great accuracy (CC 0.92 and RMSE 0.13) [13]. Similarly, Gupta et al. noted the strength of LSTM in temporal patterns but highlighted its high cost and complexity [14].

This work employs the boost approach model for time-series wave forecasting, specifically proposing the Catboost model. The model represents an enhancement over the renowned XGBoost model. CatBoost is utilised for time series forecasting because it can efficiently manage extensive datasets, mitigate overfitting risks, and deliver superior performance amidst fluctuation and complexity in waveform data [9]. In the subsequent part, we succinctly delineated versions of boosting models, including Adaboost, XGBoost, and CatBoost.

B. Adaptive Boosting(AdaBoost)

AdaBoost is based on ensemble learning; hence, it enhances the accuracy of classification problems by integrating many weak classifiers to produce a single strong classifier. In 1996, Yoav Freund and Robert Schapire invented AdaBoost. The contributions of both researchers represented a significant advance in machine learning. It resulted in several modifications and enhancements in the boosting algorithms today [15]. The technique operates on the premise of iteratively modifying the weights of the training data to address classification faults. Examples that are difficult to categorise receive greater emphasis, prompting the algorithm to enhance the prediction accuracy for these instances. This strategy improves the model's overall efficacy in classifying tasks [16]. The architecture of the AdaBoost is illustrated in Figure 1.

In 2022, Wang et al. proposed a control technique utilizing AdaBoost to improve the efficiency of an electro-hydraulic servo system. This methodology effectively enhanced tracking performance, responsiveness to sinusoidal signals, and resilience to shocks by integrating AdaBoost and RBF neural networks. The peak response frequency was 15 Hz, indicating substantial system stability and performance enhancements. [17]. In that same year, Wang et al. devised a BP-AdaBoost algorithm for diagnosing faults in aircraft wiring. By integrating BP and AdaBoost neural networks, this technique enhanced diagnostic accuracy for defects, including short circuits, open circuits, and insulation failures. It substantially lowered the misclassification rate [18].

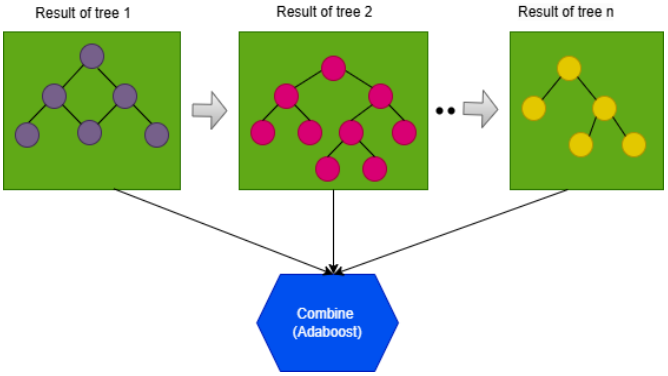


Fig. 1. The Architecture of AdaBoost

C. Extreme Gradient Boosting(XGBoost)

XGBoost (Extreme Gradient Boosting) was created by Tianqi Chen in 2014 [19]. Existing boosting techniques enhance model performance by the aggregation of many decision trees. XGBoost fundamentally aims to improve ensemble learning performance with a gradient-boosting methodology. This will be accomplished by regularization, pruning, and caching, enhancing its efficiency and efficacy in addressing various difficulties in machine learning classification and regression. The XGBoost offers several advantages in machine learning, including improved performance through highly accurate predictions, elimination of bias, assessment of feature relevance, and versatility in managing diverse machine learning tasks. The XGBoost is employed in several applications, including wave prediction, and it demonstrates efficacy in managing unbalanced datasets. The model's advantages are in classification and regression applications, where speed and accuracy are paramount [20].

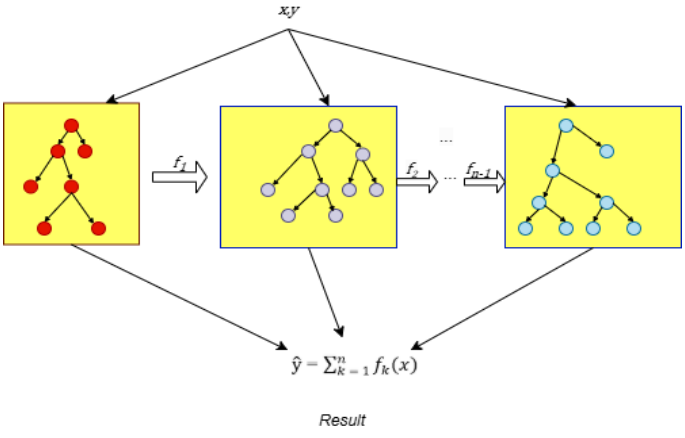


Fig. 2. The Architecture of XGBoost

The architecture of the XGboost is illustrated in Figure 2. Each example x_i It is predicted using a series of decision trees. The prediction, for example, i, i.e. $\hat{y}_i$, is calculated as the sum of the outputs of all n regression trees:

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.

$$\hat{y}_i = \sum_{K=1}^{n} f_k(x_i), \quad \{f_k \in F\}, \tag{1}$$

Here, f_k is the function represents the k decision tree, and $\boldsymbol{F}$ is the space of all regression trees. The objective of XGBoost training is to minimize the objective function, which consists of two parts:

$$\text{Obj}(\theta) = L(\theta) + \Omega(\theta), \tag{2}$$

Where $L(\theta)$ is the loss function, and $\Omega(\theta)$ is the model complexity. The model is then trained in an additive way. This means reduction iteration at the t , $\hat{y}_i^{(t)}$ is obtained by adding the prediction from the new tree $f_t(x_i)$ To the existing prediction from the previous iteration:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \tag{3}$$

At each iteration, the objective function being minimized is:

$$Obj^{(t)} = \sum_{i=1}^{n} L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \tag{4}$$

XGBoost employs second-degree estimation to calculate the loss function using first—and second-gradient statistics for faster optimization. (g_i *and* h_i) :

$$Obj^{(t)} \approx \sum_{i=1}^{n} (g_i f_t(x_i) + \frac{1}{2} h_i f t^2(x_i)) + \Omega(f_t) \tag{5}$$

here (g_i and h_i) represent the first and second gradients of the loss function about the prediction at x_i.

D. Categorical Boosting (CatBoost)

CatBoost was created by the Yandex team and was initially released to the public in 2017. CatBoost aims to enhance performance by integrating a boosting methodology with advances focused on managing categorical data, regularization, and mitigating data loss. CatBoost possesses a competitive edge compared to alternative machine learning techniques: It functions 30 to 60 times more rapidly than XGBoost and AdaBoost. CatBoost surpasses other modelling methods in accuracy and speed. [21]. It surpasses machine learning methods in prediction accuracy, such as XGBoost and AdaBoost. The model has a high level of accuracy, shown by an R2 value approaching one and a low RMSE value. This indicates that the absence of overfitting or underfitting enables the model to generalize effectively to novel data, which is essential for dependable predictions in real-world applications[22]. CatBoost is advantageous for tiny data sets since it provides rapid model fitting, execution, and prediction times relative to other machine-learning techniques.

It is assumed that the observational dataset consists of pairs $S = \{(X_i,Y_i),(X_2,Y_2),\ldots,(X_n,Y_n)\}$where $X_i = \{x_i^1, x_i^2, \ldots x_i^m\}$ is an m-dimensional vector of numerical and categorical features, an Y_i Is the value. The CatBoost model converts all numerical features to binary using the 'tree of oblivious' method, a significant predictor for floating-point binarisation, statistical analysis, and one-hot coding.

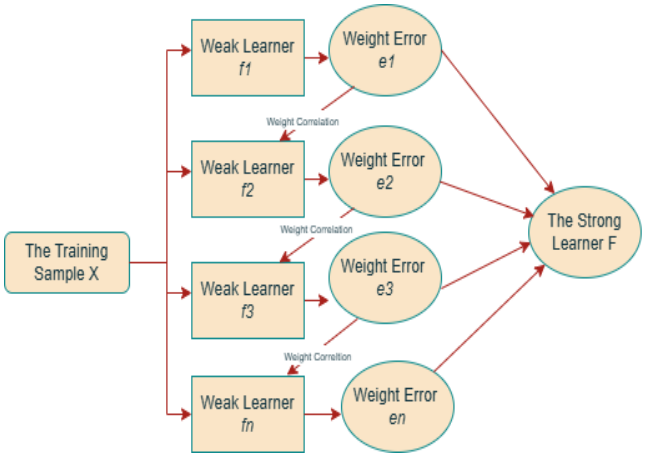


Fig. 3. The Architecture of CatBoost

The model translates category information into numerical values via two primary steps:

1. Randomize Values: The observed values are randomized to produce several sequences. This seeks to eliminate the influence of data sequence on the model's efficacy, potentially mitigating overfitting.

2. Calculation of Average Value: The goal mean value of a dataset is employed to classify categorical characteristics using the following formula:

$$x_i^k = \frac{\sum_{j=1}^{n} [x_i^k = x_i^k] \cdot Y_j}{\sum_{j=1}^{n} [x_j^k = x_i^k]} \tag{6}$$

where, $[x_j^k = x^k]$*It is* an indicator that takes the value one if x_j^k is equal to x^k And zero otherwise. This results in a value that is a weighted average of the targets based on the presence of categorical values.

Assuming that θ is $(\theta_1, \theta_2, \ldots, \theta_n)$ The values of the categorical features are converted into numerical values:

$$x_i^k = \frac{\sum_{j=1}^{p-1} [x_{\theta j}^k = x_{\theta p}^k] \cdot Y_{\sigma j} + a \cdot P}{\sum_{j=1}^{n} [x_{\theta j}^k = x_{\theta p}^k] + a} \tag{7}$$

A preceding value P and a parameter a (where $a > 0$) are included as a starting weight that aids in mitigating noise from the low-frequency range. Ultimately, concerning feature fusion, CatBoost employs an assertive methodology: 1) The initial layer of the tree performs no merging functions; 2) The subsequent layer amalgamates all combinations and categorization attributes. The architecture of the Catboost is illustrated in Figure 3. All classification attributes are included in the tree and the complete dataset. The merged categorical characteristics are transformed into numerical features; each split chosen in the tree is regarded as a categorical feature.

E. Performance Matrix

This paper uses two performance metrics: the RMSE (Root Mean Square Error) and R^2 (coefficient of determination). The RMSE quantifies the proximity of the forecast to the actual value by computing the mean of the squared differences between anticipated and observed values, followed by taking the square root of this average. The formula is provided below:

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (8)$$

Where n is the number of samples, y_i is the actual value, and $\hat{y}_i$ Is the model prediction. The formula for R^2 (coefficient of determination) that tells us to what extent the regression model explains the variability in the data:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \quad (9)$$

Here, y_i represents the actual observed value or the accurate data while $\hat{y}_i$ Denotes the value predicted by the model. $\bar{y}$ Is the mean of the y_i values. The range of R^2 Spans from 0 and 1, with values closer to 1, indicate better model performances.

III. METHODOLOGY

This study advocates creating a high-resolution wave prediction system for coastal regions with intricate geometries, utilising machine learning-based time series forecasting. A tiered numerical wave simulation generates the high-resolution wave dataset that models wave propagation from the global to the local domain. The acquired data was utilised to train machine learning models, specifically CatBoost, to forecast local wave conditions based on global wave data. The process consists of two stages: a numerical simulation that generates wave datasets and the application of machine learning to these datasets to develop high-resolution forecasting models. This research employs time series forecasting, using historical wave height (hs) data as input for the machine learning model. The data, derived from simulation results using the SWAN model, is free of missing values due to reconstruction through simulation. The methodology of this investigation is illustrated in Figure 4.

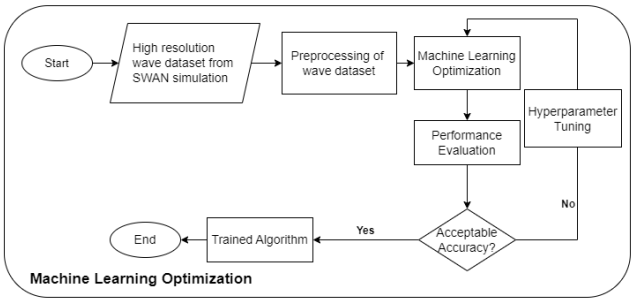


Fig. 4. Flowchart of wave height time series forecasting

A. Hyperparameter Tuning

We tuned all parameters in the model to achieve optimal performance outcomes. The GridSearch method was employed to choose the optimal parameter set. This method assesses many parameter values to identify the optimal combination. The model will be selected for data forecasting if its performance, after hyperparameter tweaking, is above that of the model without tuning.

B. Wave Height Data

This research employed wave data from Pacitan, East Java, Indonesia. An area located on the southern shore of Java Island, with coordinates between 7°55' and 8°17' South latitude and 110°55' and 111°25' East longitude. The wave data is acquired by high-resolution wave downscaling utilizing the SWAN wave model [11]. The simulation has been conducted continuously for over 9 years, from 2015 to 2024, over three nested domains, employing the identical parameters outlined in the study by Adytia et al. [1]. Figure 5 illustrates the smallest domain in the coastal region of Pacitan. This study acquired wave data from SWAN simulation outcomes, including a 1.5-hour temporal grid. The data spans from January 1, 2015, to January 15, 2024. The dataset was divided into training and testing, with 80% designated for training and 20% for testing.

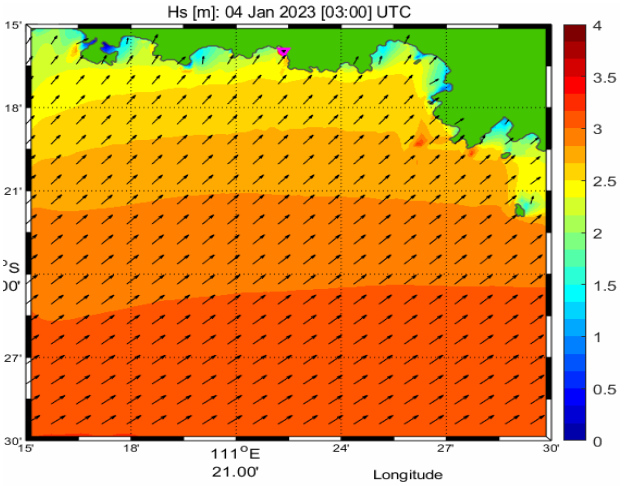


Fig. 5. Domain of simulation on the coast of Pacitan, East Java, Indonesia.

IV. RESULT AND ANALYSIS

A. Training Data Length Sensitivity

This study uses time series forecasting of wave height data from the wave height monitoring station in Pacitan, East Java, Indonesia. The training data is categorized into three temporal scenarios: a one-year scenario (January 1, 2022, to December 31, 2022), a two-year scenario (January 1, 2021, to December 31, 2022), and an eight-year scenario (January 1, 2015, to December 31, 2022). Every scenario was utilised to train the algorithm and produce forecasts for the subsequent 14 days.

TABLE I. TRAINING DATA LENGTH PERFORMANCE SENSITIVITY

Length of Training Data Catboost	Performance Matrix	
	RMSE	R^2
1 Year	0.0201	0.9965
2 Years	0.0130	0.9984
8 Years	**0.0121**	**0.9986**

Table I illustrates that the assessment findings from the 8-year scenario achieved the maximum accuracy, with an R^2 of 0.9986 and RMSE of 0.0121. The CatBoost model may utilise 8 years of training data for input time series forecasting.

B. Prediction Length Sensitivity

The length of the time series forecasting prediction period significantly influences the results' precision. A sensitivity study of the CatBoost model's performance is performed by evaluating several prediction periods: 1, 3, 5, and 7 days. In each predictive scenario, CatBoost's performance is assessed against two additional models, AdaBoost and XGBoost.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.

TABLE II. RESULT OF 1 TO 14 DAYS OF FORECASTING USING VARIOUS MODEL

Days	CatBoost		AdaBoost		XGBoost	
	R^2	RMSE	R^2	RMSE	R^2	RMSE
1	0.9638	0.0138	0.9201	0.0234	0.9587	0.0158
5	0.9934	0.0149	0.9808	0.0236	0.9895	0.0177
7	0.9979	0.0141	0.9917	0.0234	0.9962	0.0172
14	**0.9986**	**0.0121**	0.9925	0.0225	0.9975	0.0155

The performance comparison results of the three techniques are analysed to determine the most successful model, with specifications detailed in Table II. Table II indicates that CatBoost performs optimally for 14-day predictions, with an RMSE of 0.0121 and an R^2 of 0.9986. XGBoost exhibits an RMSE of 0.0155 and an R^2 of 0.9975, while AdaBoost has an RMSE of 0.0225 and an R^2 of 0.9925. CatBoost has superior performance in accuracy and consistency, as seen in Figure 6, where it yields more steady predicting results compared to XGBoost and AdaBoost, which exhibit a decline in accuracy at specific time intervals.

Figure 6. illustrates the inadequate performance of the XGBoost and AdaBoost models in forecasting data at the peak and valley points. The XGBoost and AdaBoost models underestimated most data in the peaks and valleys. Conversely, the CatBoost model offers a more accurate estimation of the peak and trough positions. It should be emphasised that the wave height variations seen are not anomalies but rather influenced by cyclonic activity in the southern Indian Ocean. This cyclone naturally affects the wave pattern at Pacitan Beach, and the pattern is well represented by CatBoost, reflecting the actual environmental phenomenon.

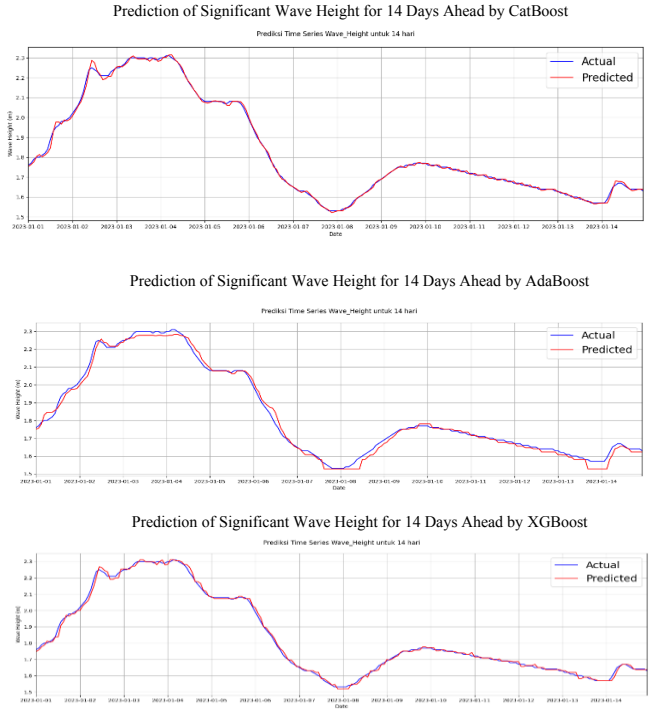


Fig 6. The 14-day prediction result for CatBoost (top), AdaBoost (middle) and XGBoost (bottom).

C. Catboost Time Series Model

The predictive outcomes indicate that the CatBoost model excels in the 14-day forecast, achieving the minimum RMSE of 0.0121 and a maximum R^2 of 0.9986. Compared to forecasts for other intervals, the 1-day forecast exhibits inferior performance, evidenced by the greatest RMSE of 0.0138 and the lowest R^2 of 0.9638. Simultaneously, the 5- and 7-day forecasts yield superior outcomes compared to the 1-day projection. However, they remain less precise than the 14-day forecast. Consequently, the 14-day forecast with CatBoost is regarded as the most accurate model for predicting wave height.

TABLE III. COMPARISON OF WAVE HEIGHT FORECASTING MODELS USING CATBOOST XGBOOST AND ADABOOST FOR 14 DAYS

Model	Default Parameter		Hyperparameter Tuning	
	RMSE	R^2	RMSE	R^2
Catboost	0.0124	0.9985	**0.0121**	**0.9986**
XGBoost	0.0174	0.9969	0.0155	0.9975
AdaBoost	0.0270	0.9853	0.0225	0.9925

In Table III, hyperparameter tuning provides improved performance on all models. CatBoost demonstrated enhanced accuracy, evidenced by a reduction in RMSE from 0.0124 to 0.0121 and an elevation in R^2 from 0.9985 to 0.9986, establishing it as the superior model. XGBoost demonstrated enhancement, as the RMSE diminished from 0.0174 to 0.0155, while the R^2 value ascended from 0.9969 to 0.9975. Concurrently, AdaBoost exhibited notable enhancement, as RMSE reduced from 0.0270 to 0.0225 and R^2 increased from 0.9853 to 0.9925. Despite the upgrades, AdaBoost's efficacy continues to be subpar compared to CatBoost and XGBoost. For example, in this study, CatBoost uses an ordered enhancement technique that prevents overfitting and captures complex patterns, including extreme fluctuations such as cyclones. The predictions generated by CatBoost almost match the actual data, indicating high accuracy. In summary, CatBoost with hyperparameter optimisation emerged as the superior model in this investigation, exhibiting the most fantastic accuracy and dependable prediction capability.

D. Optimization Parameters

TABLE IV. PARAMETERS MODELS

Model	Parameters	Best Parameters
CatBoost	Iterations = [500, 750, 1000] Learning_rate = [0.2, 0.1, 0.001, 0.0001, 0.00001] Depth = [4, 6, 8] l2_leaf_reg = [3, 5, 10]	Iterations = 1000 Learning_rate = 0.2 Depth = 4 l2_leaf_reg = 5
AdaBoost	n_estimators = [50, 100, 200, 300,400] learning_rate = [0.1, 0.3, 0.05]	n_estimators = 300 learning_rate = 0.3
XGBoost	n_estimators = [50, 100] learning_rate = [0.5, 0.1] max_depth = [1,2]	n_estimators = 100 learning_rate = 0.5 max_depth = 2

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.

Hyperparameter adjustments were made using the GridSearch technique to optimize the performance of each model. This approach seeks to identify the combination of parameters that yields the best performance. Table IV displays the hyperparameter tuning results for the three models evaluated. The optimal parameters for the CatBoost model are 1000 iterations, learning rate 0.2, depth 4, and l2_leaf_reg regularization 5. The optimal configuration for the AdaBoost model is 300 n_estimators with a learning rate of 0.3. For the XGBoost model, the optimal parameters are 100 n_estimators, learning rate 0.5, and max_depth 2.

V. CONCLUSION

This work presents wave height forecasting with CatBoost, XGBoost, and AdaBoost. The wave data utilised in this experiment was sourced from Pacitan, Indonesia. Over 9 years, 8 years of data were used for training and 1 year for testing. The experiment demonstrates that the accuracy of predictions often diminishes for long-term forecasts. The optimal prediction outcome for the CatBoost model was seen on day 14, with an RMSE of 0.0121 and an R^2 of 0.9986. Simultaneously, the XGBoost model achieved the best accuracy on day 14, with an RMSE of 0.0155 and a R^2 of 0.9975. Simultaneously, the AdaBoost model demonstrated the highest accuracy on day 14, with an RMSE of 0.0225 and an R^2 of 0.9925. The CatBoost model yields more precise predicting results than XGBoost and AdaBoost over 14 days. Based on the results of this study, the CatBoost model performs well in capturing complex patterns, including cyclonic events. However, further research is needed to confirm its adaptability in other coastal areas or extreme weather conditions. Future research could explore using the model in extreme weather conditions with higher variability and integrating it with real-time environmental monitoring systems to support predictive applications in dynamic scenarios.

REFERENCES

[1] D. Adytia *et al.*, "Modelling of Deep Learning-Based Downscaling for Wave Forecasting in Coastal Area," *Water (Switzerland)*, vol. 15, no. 1, 2023, doi: 10.3390/w15010204.

[2] F. Anggraeni, D. Adytia, and A. W. Ramadhan, "Forecasting of Wave Height Time Series Using AdaBoost and XGBoost, Case Study in Pangandaran, Indonesia," 2021 International Conference on Data Science and Its Applications, ICoDSA 2021, pp. 97– 101, 2021, doi: 10.1109/ICODSA53588.2021.9617524.

[3] Q. Guodong and C. Zhongxian, "A Comparison of Ocean Wave Height Forecasting Methods for Ocean Wave Energy Conversion Systems," *Water (Switzerland)*, vol. 15, no. 18, Sep. 2023, doi: 10.3390/w15183256.

[4] X. Feng, G. Ma, S.-F. Su, C. Huang, P. Xue, and M. K. Boswell, "A Multi-layer perceptron approach for accelerated wave forecasting in Lake Michigan," 2020.

[5] O. Hornyák and L. B. Iantovics, "AdaBoost Algorithm Could Lead to Weak Results for Data with Certain Characteristics," *Mathematics*, vol. 11, no. 8, Apr. 2023, doi: 10.3390/math11081801.

[6] H. Hu, A. J. van der Westhuysen, P. Chu, and A. Fujisaki-Manome, "Predicting Lake Erie wave heights and periods using XGBoost and LSTM," *Ocean Model (Oxf)*, vol. 164, 2021, doi: 10.1016/j.ocemod.2021.101832.

[7] S. Elsworth and S. Güttel, "Time Series Forecasting Using LSTM Networks: A Symbolic Approach," Mar. 2020, [Online]. Available: http://arxiv.org/abs/2003.05672

[8] R. Z. Zhang, X. J. Jia, and Q. F. Qian, "Seasonal forecasts of Eurasian summer heat wave frequency," *Environ Res Commun*, vol. 4, no. 2, 2022, doi: 10.1088/2515-7620/ac5364.

[9] K. Hancock, J. T. Hancock, and T. M. Khoshgoftaar, "CatBoost for Big Data: an Interdisciplinary Review." [Online]. Available: https://www.csie.ntu.edu.tw/

[10] D. Adytia, D. Saepudin, S. R. Pudjaprasetya, S. Husrin, and A. Sopaheluwakan, "A Deep Learning Approach for Wave Forecasting Based on a Spatially Correlated Wind Feature, with a Case Study in the Java Sea, Indonesia," *Fluids*, vol. 7, no. 1, Jan. 2022, doi: 10.3390/fluids7010039.

[11] L. Holthuijsen, N. Booij, R. C. Ris, and L. H. Holthuijsen, "A third-generation wave model for coastal regions, Part I, Model description and validation A third-generation wave model for coastal regions 1. Model description and validation," 1999. [Online]. Available: https://www.researchgate.net/publication/215722037

[12] S. Fan, N. Xiao, and S. Dong, "A novel model to predict significant wave height based on long short-term memory network," *Ocean Engineering*, vol. 205, Jun. 2020, doi: 10.1016/j.oceaneng.2020.107298.

[13] V. J. Didit and A. Adiwijaya, "Wave Height Prediction based on Wind Information by using General Regression Neural Network, study case in Jakarta Bay."

[14] K. K. Gupta, A. Anil, P. R. Kaveri, and A. Anand, "Understanding the need for machine learning as a solution for financial analysis of IT industries," in 2020 International Conference on Computer Communication and Informatics (ICCCI), Coimbatore, India, Jan. 2020, pp. 1–6. DOI: 10.1109/ICCCI48352.2020.9104058.

[15] Y. Ding, H. Zhu, R. Chen, and R. Li, "An Efficient AdaBoost Algorithm with the Multiple Thresholds Classification," *Applied Sciences (Switzerland)*, vol. 12, no. 12, Jun. 2022, doi: 10.3390/app12125872.

[16] K. W. Walker and Z. Jiang, "Application of adaptive boosting (AdaBoost) in demand-driven acquisition (DDA) prediction: A machine-learning approach," *Journal of Academic Librarianship*, vol. 45, no. 3, pp. 203–212, May 2019, doi: 10.1016/j.acalib.2019.02.013.

[17] X. Wang, G. Shi, M. Liu, and Y. Feng, "Research on robust control of continuous rotary electro-hydraulic servo motor based on AdaBoost prediction," *Measurement and Control (United Kingdom)*, 2022, doi: 10.1177/00202940221089241.

[18] J. K. Tsai and C. H. Hung, "Improving Adaboost classifier to predict enterprise performance after covid-19," *Mathematics*, vol. 9, no. 18, Sep. 2021, doi: 10.3390/math9182215.

[19] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Association for Computing Machinery, Aug. 2016, pp. 785–794. doi: 10.1145/2939672.2939785.

[20] Y. Wang, Z. Pan, J. Zheng, L. Qian, and M. Li, "A Hybrid Ensemble method for Pulsar Candidate Classification," Sep. 2019, doi: 10.1007/s10509-019-3602-4.

[21] P. Kumar *et al.*, "Evaluation of Catboost method for predicting weekly pan evaporation: Case study of subtropical and subhumid regions of India," 2022, doi: 10.21203/rs.3.rs-1538970/v1.

[22] M. M. Muhammed, A. A. Ibrahim, R. L. Ridwan, R. O. Abdulaziz, and G. A. Saheed, "Comparison of the CatBoost Classifier with other Machine Learning Methods," 2020. [Online]. Available: https://www.researchgate.net/publication/348277609

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:09:29 UTC from IEEE Xplore. Restrictions apply.